TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	0.0 ± 0.00
Correlation Stability	78.0 ± 0.00
Distribution Consistency	0.0 ± 0.00
Entropy Preservation	0.0 ± 0.00
Mean Z Score Drift	0.0 ± 0.00
Categorical Fidelity	29.0 ± 0.00
Privacy Risk	0.0 ± 0.00
Data Density	0.0 ± 0.00
Coverage Score	0.0 ± 0.00
Feature Accuracy	0.0 ± 0.00
Novelty Score	0.0 ± 0.00

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Dataset Audit: indian_liver_patient_dataset.csv

1. Dataset Portrait

Attribute	Value
Rows	583
Columns	11
Numeric features	10
Categorical features	1
Missing cells (%)	0.1%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	gpt-4o-mini
Folds (*13)	10
Context % (*14)	0%
Generation % (*15)	100%
Join context (*16)	False

3. Statistical Significance & Data Drift

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	0.0 ± 0.00
Correlation Stability (*2)	78.0 ± 0.00
Distribution Consistency (*3)	0.0 ± 0.00
Entropy Preservation (*4)	0.0 ± 0.00
Mean Z Score Drift (*5)	0.0 ± 0.00
Categorical Fidelity (*6)	29.0 ± 0.00
Privacy Risk (*7)	0.0 ± 0.00
Data Density (*8)	0.0 ± 0.00
Coverage Score (*9)	0.0 ± 0.00
Feature Accuracy (*10)	0.0 ± 0.00
Novelty Score (*11)	0.0 ± 0.00

*[1] (*12) Algorithm: Contextual constraint or model parameter.*

*[2] (*13) Folds: Contextual constraint or model parameter.*

*[3] (*14) Context %: Contextual constraint or model parameter.*

*[4] (*15) Generation %: Contextual constraint or model parameter.*

*[5] (*16) Join context: Contextual constraint or model parameter.*

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: LR

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7034188034188034	N/A	0.676923076923077	-0.026495726495726492	
F1	0.8108734553482932	N/A	0.38171002888623495	-0.42916342646205835	*

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.558974358974359	N/A	0.5393162393162393	-0.01965811965811965	
F1	0.5587432119380917	N/A	0.4984815965127211	-0.06026161542537064	

Model: KNN

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6863247863247863	N/A	0.6675213675213676	-0.018803418803418803	
F1	0.7971299319768493	N/A	0.3799898227598656	-0.41714010921698386	*

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7068376068376068	N/A	0.7068376068376068	0	
F1	0.8279091624529537	N/A	0.2506943890174794	-0.5772147734354742	*

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6974358974358974	N/A	0.6128205128205129	-0.0846153846153846	*
F1	0.7991453026080226	N/A	0.4232035036663809 3	-0.3759417989416418	*

Model: AB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6743589743589744	N/A	0.6658119658119659	-0.008547008547008562	
F1	0.7821343746683586	N/A	0.36846794273739436	-0.41366431930964254	*

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7025641025641025	N/A	0.6461538461538463	-0.0564102564102564	*
F1	0.7997903571549948	N/A	0.4502709684995780 4	-0.3495193886554167	*

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6837606837606838	N/A	0.6555555555555556	-0.028205128205128206	
F1	0.789863487125633	N/A	0.4271755442676052	-0.3626879428580278	*

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6581196581196582	N/A	0.5965811965811966	-0.06153846153846153	*
F1	0.7613799944274808	N/A	0.457392701385838	-0.3039872930416428	*

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7042735042735042	N/A	0.6623931623931625	-0.04188034188034188	*
F1	0.8080874637681378	N/A	0.3786295479243847	-0.42945791584375304	*

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	65	Female	0.7	0.1	187	16	18	6.8	3.3	0.9
2	62	Male	10.9	5.5	699	64	100	7.5	3.2	0.74
3	62	Male	7.3	4.1	490	60	68	7.0	3.3	0.89
4	58	Male	1.0	0.4	182	14	20	6.8	3.4	1.0
5	72	Male	3.9	2.0	195	27	59	7.3	2.4	0.4
6	46	Male	1.8	0.7	208	19	14	7.6	4.4	1.3
7	26	Female	0.9	0.2	154	16	12	7.0	3.5	1.0
8	29	Female	0.9	0.3	202	14	11	6.7	3.6	1.1
9	17	Male	0.9	0.3	202	22	19	7.4	4.1	1.2
10	55	Male	0.7	0.2	290	53	58	6.8	3.4	1.0
11	57	Male	0.6	0.1	210	51	59	5.9	2.7	0.8
12	72	Male	2.7	1.3	260	31	56	7.4	3.0	0.6
13	64	Male	0.9	0.3	310	61	58	7.0	3.4	0.9
14	74	Female	1.1	0.4	214	22	30	8.1	4.1	1.0
15	61	Male	0.7	0.2	145	53	41	5.8	2.7	0.87
16	25	Male	0.6	0.1	183	91	53	5.5	2.3	0.7
17	38	Male	1.8	0.8	342	168	441	7.6	4.4	1.3
18	33	Male	1.6	0.5	165	15	23	7.3	3.5	0.92
19	40	Female	0.9	0.3	293	232	245	6.8	3.1	0.8
20	40	Female	0.9	0.3	293	232	245	6.8	3.1	0.8

B.1 Audit Trace: Fold 1 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	33	Male	7.1	3.7	196	622	497	6.9	3.6	1.09
2	38	Male	2.2	1.0	310	119	42	7.9	4.1	1.0
3	64	Male	3.0	1.4	248	46	40	6.5	3.2	0.9
4	40	Female	0.9	0.2	285	32	27	7.7	3.5	0.8

B.2 Audit Trace: Fold 2 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	50	Female	1.0	0.3	191	22	31	7.8	4.0	1.0
2	52	Female	0.6	0.1	194	10	12	6.9	3.3	0.9
3	30	Male	0.8	0.2	182	46	57	7.8	4.3	1.2
4	58	Male	0.8	0.2	298	33	59	6.2	3.1	1.0

B.3 Audit Trace: Fold 3 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	44	Female	1.9	0.6	298	378	602	6.6	3.3	1.0
2	30	Male	0.8	0.2	182	46	57	7.8	4.3	1.2
3	53	Male	0.9	0.4	238	17	14	6.6	2.9	0.8
4	28	Male	0.8	0.3	190	20	14	4.1	2.4	1.4

B.4 Audit Trace: Fold 4 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	42	Male	0.7	0.2	197	64	33	5.8	2.4	0.7
2	69	Male	0.9	0.2	215	32	24	6.9	3.0	0.7
3	53	Male	1.6	0.9	178	44	59	6.5	3.9	1.5
4	49	Male	1.3	0.4	206	30	25	6.0	3.1	1.06

B.5 Audit Trace: Fold 5 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	33	Male	2.1	0.7	205	50	38	6.8	3.0	0.7
2	22	Female	6.7	3.2	850	154	248	6.2	2.8	0.8
3	48	Male	5.8	2.5	802	133	88	6.0	2.8	0.8
4	84	Female	0.7	0.2	188	13	21	6.0	3.2	1.1

B.6 Audit Trace: Fold 6 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	50	Male	0.9	0.3	194	190	73	7.5	3.9	1.0
2	49	Male	0.6	0.1	218	50	53	5.0	2.4	0.9
3	18	Male	0.8	0.2	228	55	54	6.9	4.0	1.3
4	30	Female	0.7	0.2	194	32	36	7.5	3.6	0.92

B.7 Audit Trace: Fold 7 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	13	Female	0.7	0.2	350	17	24	7.4	4.0	1.1
2	54	Female	23.2	12.6	574	43	47	7.2	3.5	0.9
3	63	Male	0.9	0.2	194	52	45	6.0	3.9	1.85
4	46	Male	0.6	0.2	290	26	21	6.0	3.0	1.0

B.8 Audit Trace: Fold 8 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	48	Male	0.7	0.2	208	15	30	4.6	2.1	0.8
2	60	Male	22.8	12.6	962	53	41	6.9	3.3	0.9
3	27	Male	1.0	0.3	180	56	111	6.8	3.9	1.85
4	28	Female	0.8	0.2	309	55	23	6.8	4.1	1.51

B.9 Audit Trace: Fold 9 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	26	Male	2.0	0.9	195	24	65	7.8	4.3	1.2
2	47	Male	2.7	1.3	275	123	73	6.2	3.3	1.1
3	74	Male	1.0	0.3	175	30	32	6.4	3.4	1.1
4	48	Male	3.2	1.6	257	33	116	5.7	2.2	0.62

B.10 Audit Trace: Fold 10 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	54	Female	5.5	3.2	350	67	42	7.0	3.2	0.8
2	43	Male	22.5	11.8	143	22	143	6.6	2.1	0.46
3	57	Male	4.5	2.3	315	120	105	7.0	4.0	1.3
4	65	Male	0.7	0.2	199	19	22	6.3	3.6	1.3